

Technical Backend Document

Human Resources Module — Fiscal Payroll Calculations

Priority Jurisdictions: Mexico, Canada, Colombia, Brazil

Product: Indice ERP | Module: Human Resources | Section: Payroll

Backend implementation guide. Not legal, tax, or accounting advice. Final rates and tables must be loaded from official sources and validated by local payroll advisors before production use.

1. Executive Summary

This document defines backend calculation instructions for fiscal payroll in the four priority jurisdictions for Indice: Mexico, Canada, Colombia, and Brazil. The current frontend already shows the operational structure needed for payroll review: payroll runs, simplified vs detailed payroll, statutory/internal-only toggles, employee lines, variable pay, deductions, employer cost summary, and country-specific concepts. However, payroll calculations must not live in React. Backend must become the single calculation authority.

The correct architecture is not one table with more columns. It is Payroll Core plus jurisdiction calculation adapters. Each adapter owns statutory bases, employee deductions, employer obligations, validations, component codes, and effective-dated rule versions.

Decision	Backend Requirement
Single source of truth	Backend computes all fiscal payroll amounts. Frontend only captures approved inputs and displays backend results.
Jurisdiction model	Resolve payroll by legal entity, country, subdivision/province/state, currency, period, worker profile, and rule version.
No mixed statutory runs	A payroll run must not mix countries, currencies, legal entities, pay schedules, or incompatible subdivisions.
Immutable approvals	Approved payroll runs must store input snapshots, rule snapshots, component breakdowns, audit events, and export references.
Frontend migration	Remove all hardcoded statutory rates and country formulas from React. Replace with backend preview and run detail payloads.

2. Current Frontend/Repo Findings

The uploaded repository shows a transitional payroll implementation. The UI and API contract already expose enough concepts to guide backend architecture, but current payroll preferences remain Mexico-shaped percentage fields and cannot safely support Canada, Colombia, and Brazil.

Area	Current Finding	Backend Implication
Payroll preferences	Fields include <code>isr_rate</code> , <code>imss_user_rate</code> , <code>infonavit_user_rate</code> , <code>imss_employer_rate</code> , <code>infonavit_employer_rate</code> , <code>sar_employer_rate</code> .	These fields are transitional only. They must be demoted to company defaults or replaced by effective-dated country rule packs.
Payroll run lines	Lines include salary type, base salary, hourly rate, days payable, leave days, absence days, regular hours, overtime hours, <code>include_in_fiscal</code> , gross, deductions, employer contributions, net.	This is a good universal payroll line foundation. Add jurisdiction, currency, rule snapshot, worker profile snapshot, and component-level calculation details.
Line items	Items support category: earning, deduction, employer_contribution; source_type: computed, manual, computed_tax.	Keep this model, but standardize component codes per jurisdiction and preserve calculation metadata.
Frontend API	React expects list, create, detail, update line, process, approve, mark-paid, cancel.	Keep lifecycle. Add preview-first API and immutable snapshots.
Globalization plan	Existing docs already recommend payroll catalog, legal entity profiles, worker profiles, rule versions, preview, snapshots, country packs.	Use that architecture as vNext backend target.

3. Backend Core Principle

Backend must calculate every official payroll value. Frontend must not calculate taxes, social security, employer obligations, or statutory benefits. React can calculate only harmless display counts or local UI totals when explicitly marked as non-authoritative.

Must Be Backend-Calculated	Examples
Gross earnings	Period salary, hourly wages, overtime, bonuses, commissions, taxable and non-taxable earnings.
Employee deductions	Income tax withholding, social security employee share, housing deductions, pension, loans, statutory withholdings.
Employer obligations	Employer social security, payroll taxes, funds, parafiscals, FGTS, CPP/EI employer, ARL, benefits provisions.
Net pay	Gross earnings minus employee deductions.
Total employer cost	Gross earnings plus employer obligations and provisions.
Compliance output	Payslip breakdowns, export components, tax/reporting readiness flags.

4. Payroll Run Lifecycle

State	Description	Editable
Draft	Payroll run exists but can be recalculated. Inputs and manual adjustments may still change.	Yes
Processed	Backend calculation has been executed and review is expected.	Limited
Approved	Run is approved. Snapshot is locked. No recalculation without explicit reopen/reversal.	No
Paid	Payment completed. No calculation changes allowed.	No
Cancelled	Run cancelled. Must remain auditable.	No

Required flow:

1. Resolve legal entity, country, subdivision, currency, pay calendar, and rule version.
2. Load eligible workers from payroll worker profiles, not only HR employee records.
3. Aggregate attendance and schedule inputs from Control daily records/events.
4. Aggregate compensation inputs: salary, hourly rate, recurring earnings/deductions, manual adjustments.
5. Execute jurisdiction adapter and produce preview components.
6. Persist draft with input snapshots and rule snapshots.
7. Allow limited audited adjustments in Draft/Processed only.
8. Approve and lock calculations.
9. Export payslips, summary, accounting journal, and jurisdiction-specific files from snapshots.

5. Recommended Backend Data Model

Table / Object	Purpose
payroll_supported_countries	Country and subdivision catalog: MX, CA, CA-QC, CO, BR; future US, etc.
payroll_rule_versions	Effective-dated statutory rules, rates, brackets, ceilings, minimums, formulas, and component metadata.
payroll_legal_entities	Employer entity used for payroll reporting: country, currency, registrations, default calendar.
payroll_worker_profiles	Worker payroll identity, jurisdiction, tax identifiers, compensation mode, contribution settings.
payroll_worker_compensation_items	Recurring earnings/deductions: bonuses, commissions, benefits, loans, allowances.
payroll_runs	Run header: legal entity, country, subdivision, currency, period, status, totals.
payroll_run_lines	Per-worker frozen calculation result.
payroll_run_line_items	Component-level earnings, deductions, employer obligations.
payroll_run_input_snapshots	Frozen attendance, worker profile, salary, rates, and manual inputs used for calculation.
payroll_run_rule_snapshots	Exact rule version and resolved parameters used by the adapter.
payroll_run_events	Audit trail: preview, calculate, update input, process, approve, pay, cancel, reopen.
payroll_exports	Generated PDF/CSV/payslip/accounting/export metadata.

6. Universal Calculation Contract

```
PayrollRunPreviewRequest {
  legal_entity_id
  country_code           // MX, CA, CO, BR
  subdivision_code       // CA-ON, CA-QC, MX state, CO dept if needed, BR state if needed
  currency               // MXN, CAD, COP, BRL
  pay_period             // weekly, biweekly, monthly
  period_start_date
  period_end_date
  grouping_mode          // single, unit, business; later legal_entity first
  worker_scope_filters
}

PayrollRunLineResult {
  worker_profile_snapshot
  attendance_input_snapshot
  rule_version_snapshot
  earnings_components[]
  employee_deduction_components[]
  employer_obligation_components[]
  provisions_components[]
  gross_pay
  total_employee_deductions
  net_pay
  total_employer_obligations
  total_payroll_cost
  warnings[]
  blocking_errors[]
}
```

Formula	Definition
period_salary	daily_salary * paid_days OR salary_period_amount prorated by pay calendar.
hourly_regular_pay	regular_hours * hourly_rate.
overtime_pay	sum(overtime_hours_by_category * overtime_rate_multiplier * hourly_rate).
variable_pay	sum(active bonuses + commissions + positive adjustments that affect pay).
gross_pay	period_salary + hourly_regular_pay + overtime_pay + variable_pay + taxable_benefits + allowances.
statutory_tax_base	sum components flagged taxable for the jurisdiction.
social_security_base	sum components flagged integrates_social_security_base, capped by jurisdiction ceilings.
employee_deductions	sum statutory employee deductions + internal deductions + negative adjustments.
net_pay	gross_pay - employee_deductions.
employer_obligations	sum employer social contributions + payroll taxes + funds + provisions.
total_payroll_cost	gross_pay + employer_obligations + employer provisions if accrued.

7. Variable Concepts and Flags

Field	Required Meaning
type	BONUS, COMMISSION, ADJUSTMENT, ALLOWANCE, BENEFIT, LOAN_DEDUCTION, REIMBURSEMENT, OVERTIME, LEAVE_PAYMENT.
amount	Decimal value in run currency. BigDecimal only.
taxable	Included in income tax base for the country.
integrates_social_security_base	Included in IMSS/CPP/QPP/EPS/pension/INSS base depending on jurisdiction.
integrates_benefit_base	Included in severance, vacation, 13th salary, prima, cesantias, FGTS, etc.
apply_to_statutory_payroll	If false, concept is internal only and excluded from fiscal/statutory calculation.
affects_net_pay	If true, changes amount paid to worker. If false, only records provision/cost.
source	SYSTEM, MANUAL, IMPORT, ATTENDANCE, ADAPTER.

effective_date	Used to resolve period eligibility.
audit_required	Any manual change to a fiscal component requires audit.

8. Mexico Payroll Adapter (MX)

Mexico payroll is fiscal-compliance-first. The adapter must be prepared for CFDI payroll readiness and must treat ISR, subsidio al empleo, IMSS, INFONAVIT, SAR, state payroll tax, vacation premium, Christmas bonus, and employer obligations as backend-calculated components.

Input / Profile Field	Backend Use
rfc, curp, nss	Required identifiers for statutory payroll and CFDI readiness.
employment_type, contract_type, regime_code	Required for payroll classification.
daily_salary	Base daily salary.
integrated_daily_salary / SBC	IMSS contribution base. Must support legal caps and integration factors.
state_code	Used for state payroll tax configuration.
infonavit_credit_type	NONE, FIXED_AMOUNT, PERCENTAGE, VSM/UMA-like mode.
is_statutory_payroll	If false, skip ISR, IMSS, INFONAVIT, SAR, employer obligations and mark internal-only.

Component Code	Formula / Rule
MX_PERIOD_SALARY	period_salary = daily_salary * paid_days.
MX_OVERTIME	overtime_pay = sum(hours * hourly_rate * multiplier). Multipliers must come from rule version.
MX_VARIABLE_PAY	Sum taxable/non-taxable bonuses, commissions, and adjustments based on flags.
MX_ISR_TAXABLE_BASE	Sum active statutory taxable concepts for the period.
MX_ISR	excess = base - lower_limit; marginal_tax = excess * rate; isr_before_subsidy = fixed_fee + marginal_tax; final_isr = max(isr_before_subsidy - employment_subsidy, 0). Brackets must be effective-dated.
MX_EMPLOYMENT_SUBSIDY	Apply only when worker is statutory, jurisdiction is MX, and taxable base falls inside active subsidy rule/table.
MX_EMPLOYEE_IMSS	Sum employee IMSS components calculated from SBC, days, caps, UMA, and active rates by branch.
MX_EMPLOYER_IMSS	Sum employer IMSS components calculated from SBC, days, caps, UMA, risk class and active rates by branch.
MX_INFONAVIT_EMPLOYEE	If credit exists: fixed amount, percentage of base, or VSM/UMA-derived amount according to credit type.
MX_INFONAVIT_EMPLOYER	Employer housing contribution from SBC/base and active rate.
MX_SAR	sar = SBC * paid_days * active SAR rate.
MX_STATE_PAYROLL_TAX	payroll_tax = state_tax_base * state_rate. State rate must be configured by state and effective date.
MX_VACATION_PREMIUM	vacation_premium = vacation_days * daily_salary * premium_rate.
MX_CHRISTMAS_BONUS	christmas_bonus = daily_salary * bonus_days * worked_proportion.
MX_NET_PAY	gross_pay - ISR - employee IMSS - INFONAVIT employee - internal deductions.
MX_TOTAL_EMPLOYER_COST	gross_pay + employer IMSS + employer INFONAVIT + SAR + state payroll tax + provisions.

Validation	Rule
No fiscal identifiers	Block statutory payroll unless RFC/CURP/NSS requirements are satisfied.
Internal-only worker	Do not calculate fiscal components. Keep display and internal payment only.
Missing rule version	Block calculation. Never use silent fallback.
Manual fiscal concept change	Allow only in Draft/Processed with audit and reason.
CFDI readiness	Mark run as not export-ready if required CFDI fields are missing.

9. Canada Payroll Adapter (CA)

Canada payroll is deduction-formula and province-driven. The backend must support standard Canada and Quebec as separate subdivision behavior. CRA payroll formulas cover federal, provincial/territorial deductions except Quebec, CPP, and EI. Quebec requires QPP, QPIP, Quebec tax logic, and adjusted EI handling.

Input / Profile Field	Backend Use
sin	Required identifier for statutory payroll.
province_code	Required for provincial tax and Quebec special handling.
td1_claim_code / federal_claim_amount	Required to calculate income tax withholding.
provincial_claim_code / amount	Required for provincial/territorial tax.
pay_type	Salary, hourly, commission, bonus, pension/other if later supported.
cpp_exempt, ei_exempt	Contribution applicability flags.
qpp_exempt, qpip_exempt	Quebec applicability flags.
vacation_pay_policy	Accrued, paid each period, or paid when taken.

Component Code	Formula / Rule
CA_REGULAR_PAY	salary or hourly regular pay for the period.
CA_OVERTIME_PAY	$\text{overtime_hours} * \text{hourly_rate} * \text{active provincial/federal overtime multiplier}$.
CA_VACATION_PAY	$\text{vacationable_earnings} * \text{vacation_rate}$ OR accrued provision according to worker/legal entity policy.
CA_TAXABLE_BENEFITS	Sum taxable benefits included in income tax and/or CPP/EI base based on flags.
CA_FEDERAL_TAX	Use CRA T4127/T4032 formula tables for the pay period, annualization, claim codes and credits.
CA_PROVINCIAL_TAX	Use CRA provincial/territorial formulas except Quebec. Province rules must be effective-dated.
CA_CPP	Calculate CPP employee contribution on pensionable earnings using exemption, max pensionable earnings, rate, and annual-to-date cap.
CA_CPP2	Calculate second additional CPP where applicable using second ceiling and active rate.
CA_EI	Calculate EI employee premium on insurable earnings using max insurable earnings, rate, and annual-to-date cap.
CA_EMPLOYER_CPP	Usually matches employee CPP/PP2 amount unless configured otherwise by current rules.
CA_EMPLOYER_EI	$\text{employee_EI} * \text{employer_multiplier}$, subject to annual max.
CA_WORKERS_COMP	Optional employer obligation: insurable earnings * province/industry rate. Must be configurable.
CA_NET_PAY	$\text{gross_pay} - \text{federal tax} - \text{provincial tax} - \text{CPP/PP2} - \text{EI} - \text{other deductions}$.
CA_TOTAL_EMPLOYER_COST	$\text{gross_pay} + \text{employer CPP/PP2} + \text{employer EI} + \text{workers comp} + \text{benefits}$.

Quebec Component	Formula / Rule
CA_QC_QPP	Use QPP base/additional rates, ceilings and exemption instead of CPP for Quebec workers.
CA_QC_QPP2	Use QPP second additional contribution where applicable.
CA_QC_QPIP	Calculate QPIP employee premium on eligible earnings up to ceiling.
CA_QC_PROVINCIAL_TAX	Use Quebec-specific provincial tax rule source; do not use standard CRA provincial formulas for Quebec.
CA_QC_EMPLOYER_QPP_QPIP_EI	Employer obligations must include QPP/QPP2/QPIP and adjusted EI where applicable.

10. Colombia Payroll Adapter (CO)

Colombia payroll is prestacional and social-security-heavy. The backend must calculate employee health/pension contributions, employer health/pension/ARL, parafiscales, transport allowance, severance accrual, severance interest, service bonus, vacation provision, and withholding tax when applicable.

Input / Profile Field	Backend Use
document_id, contribution_type	Required for statutory payroll and PILA-style readiness.
salary_type, base_salary	Base compensation. Must support integral salary and standard salary.
is_integral_salary	Changes contribution and benefit treatment.
arl_risk_class	Determines ARL rate.
transport_allowance_eligible	Controls auxilio de transporte.
exoneration_flags	Employer exemptions for health/SENA/ICBF when legally applicable.
salary_components_flags	Determines IBC and benefit bases.

Component Code	Formula / Rule
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CO_BASIC_SALARY	period_salary = monthly_salary prorated by period or daily_salary * paid_days.
CO_OVERTIME_SURCHARGES	Calculate overtime/night/Sunday/holiday surcharges using configured labor multipliers.
CO_TRANSPORT_ALLOWANCE	Apply if eligible by salary threshold and period. Not always part of contribution base; treatment must be rule-driven.
CO_IBC	Contribution base for health, pension, ARL: salary components flagged as IBC-applicable, respecting minimum/maximum bases.
CO_EMPLOYEE_HEALTH	employee_health = IBC * employee_health_rate.
CO_EMPLOYEE_PENSION	employee_pension = IBC * employee_pension_rate.
CO_SOLIDARITY_FUND	Apply additional employee pension solidarity percentages when income threshold applies.
CO_WITHHOLDING_TAX	retencion_laboral = taxable_labor_base transformed to UVT and calculated by configured withholding procedure.
CO_EMPLOYER_HEALTH	employer_health = IBC * employer_health_rate unless exonerated.
CO_EMPLOYER_PENSION	employer_pension = IBC * employer_pension_rate.
CO_ARL	arl = IBC * arl_rate_by_risk_class.
CO_CCF	caja_compensacion = parafiscal_base * CCF rate.
CO_ICBF	icbf = parafiscal_base * ICBF rate unless exempt.
CO_SENA	sena = parafiscal_base * SENA rate unless exempt.
CO_CESANTIAS_PROVISION	cesantias = benefit_base * days_worked / 360.
CO_CESANTIAS_INTEREST	interest = cesantias_accrued * 12% * days_worked / 360. Rate must be configurable.
CO_SERVICE_BONUS_PROVISION	prima = benefit_base * days_worked / 360.
CO_VACATION_PROVISION	vacation = salary_base_excluding_transport * days_worked / 720.
CO_NET_PAY	gross_pay + transport_allowance - employee health - employee pension - solidarity - withholding - internal deductions.
CO_TOTAL_EMPLOYER_COST	gross_pay + employer health + employer pension + ARL + CCF + ICBF + SENA + provisions.

Validation	Rule
Missing ARL risk class	Block statutory payroll or mark as non-compliant.
Integral salary	Use special basis and benefit rules. Do not apply normal prestaciones formula blindly.
Exemptions	Employer health/SENA/ICBF exemptions must be explicit and auditable.
IBC limits	Apply minimum/maximum IBC rules from effective-dated tables.

11. Brazil Payroll Adapter (BR)

Brazil payroll is employer-obligation-heavy and eSocial-driven. The backend must calculate employee INSS, IRRF, FGTS, employer INSS, RAT, third-party contributions, vacation and constitutional one-third, 13th salary accrual/payment, transportation/benefit deductions, and eSocial component readiness.

Input / Profile Field	Backend Use
cpf, pis_pasep_nit	Required worker identifiers.
employment_type	CLT, intern, contractor, temporary; controls statutory treatment.
dependents_count	Used in IRRF deductions when applicable.
salary_base	Monthly salary base.
fgts_applicable	Controls FGTS deposit.
rat_class	Risk/work accident contribution rate.
third_party_rate_profile	System S/third-party contribution configuration.
transport_voucher_policy	Vale transporte deduction and employer benefit treatment.

Component Code	Formula / Rule
BR_MONTHLY_SALARY	monthly salary prorated by paid days.
BR_OVERTIME	overtime_hours * hourly_rate * active overtime multiplier. Include night shift/holiday rules when enabled.
BR_DSR	Weekly paid rest impact from variable pay when applicable.
BR_INSS_EMPLOYEE	Progressive employee INSS by bracket, subject to ceiling. Brackets must be effective-dated.
BR_IRRF_BASE	taxable earnings - INSS employee - dependent deductions - other legal deductions.
BR_IRRF	Progressive income tax withholding by active IRRF table.
BR_VALE_TRANSPORTE_EMPLOYEE	Employee transportation deduction according to policy and legal cap.
BR_BENEFITS_EMPLOYEE	Meal/health/other benefit deductions according to worker policy.
BR_FGTS	fgts = FGTS base * active FGTS rate. Usually employer deposit,

	not employee deduction.
BR_INSS_EMPLOYER	$\text{employer_inss} = \text{employer_social_security_base} * \text{employer rate.}$
BR_RAT	$\text{rat} = \text{risk_base} * \text{RAT rate by risk class.}$
BR_THIRD_PARTY_CONTRIBUTIONS	$\text{third_party} = \text{base} * \text{configured third-party/Sistema S rate.}$
BR_VACATION_PAY	$\text{vacation_salary} = \text{salary_base} * \text{vacation_days} / 30.$
BR_VACATION_ONE_THIRD	$\text{one_third} = \text{vacation_pay} / 3.$
BR_13TH_SALARY_PROVISION	13th_accrual = eligible_earnings / 12 for each eligible month; final payment adjusted by advances.
BR_NET_PAY	$\text{gross_pay} - \text{INSS employee} - \text{IRRF} - \text{employee benefits} - \text{internal deductions.}$
BR_TOTAL_EMPLOYER_COST	$\text{gross_pay} + \text{FGTS} + \text{employer INSS} + \text{RAT} + \text{third-party contributions} + \text{vacation/13th provisions.}$

Validation	Rule
Missing CPF	Block statutory payroll.
Missing rule version	Block calculation; Brazil cannot use generic rates.
eSocial readiness	Component codes must map to eSocial rubrics/nature codes where export is enabled.
13th salary	Keep monthly provision separate from annual/December payroll output.
Vacation	Track vacation payment and one-third separately.

12. Required Component Category Standard

Category	Examples	Backend Notes
EARNING	BASE_SALARY, HOURLY_PAY, OVERTIME, BONUS, COMMISSION, VACATION_PAY, 13TH_SALARY, TRANSPORT_ALLOWANCE.	Positive worker compensation.
EMPLOYEE_DEDUCTION	ISR, CPP, EI, IMSS, INFONAVIT, EPS, PENSION, INSS, IRRF, LOAN, BENEFIT_DEDUCTION.	Reduces net pay.
EMPLOYER_OBLIGATION	Employer CPP/EI, IMSS employer, SAR, FGTS, ARL, CCF, ICBF, SENA, RAT.	Does not reduce net pay but increases employer cost.
PROVISION	Cesantias, prima, vacation provision, 13th salary accrual.	Accruals; can be shown in employer cost if company policy requires.
INTERNAL_ONLY	Internal adjustments not included in fiscal payroll.	Must be clearly separated and not exported as statutory.

13. Recalculation Triggers

Trigger	Required Action
Attendance changes	Recalculate paid days, absences, overtime, regular hours, and attendance-derived earnings.
Salary/rate change	Recalculate gross, bases, taxes, contributions, provisions.
Worker profile change	Recalculate jurisdiction, applicability, identifiers, contribution flags.
Variable concept change	Recalculate bases and all dependent statutory components.
Tax/social rule update	New rule version only. Do not mutate approved runs.
Manual adjustment	Recalculate Draft/Processed run and write audit event.
Run scope change	Rebuild preview. Reject if it creates mixed country/currency/legal entity.

14. Precision, Rounding, and Reproducibility

Rule	Instruction
Decimal type	Use BigDecimal/decimal only. Never double/float for money.
Internal precision	At least 6 decimals during internal calculation.
Final money values	Round to 2 decimals using jurisdiction/company rounding policy.
Order of operations	Define and snapshot the calculation order per rule version.
Annual-to-date caps	Track YTD employee and employer contributions for capped systems such as CPP/EI and similar ceilings.
Approved runs	Never recalculate automatically. Use snapshot values only.

15. API Requirements

Endpoint	Purpose
GET /api/v1/hr/payroll/catalog/countries	Return supported countries, subdivisions, currency, required worker fields, and enabled rule packs.
GET/PUT /api/v1/hr/payroll/legal-entities/{id}/profile	Configure employer registration, country, currency, payroll calendar, rule pack defaults.
GET/PUT /api/v1/hr/payroll/workers/{id}/profile	Configure worker payroll profile and statutory identifiers.
POST /api/v1/hr/payroll/runs/preview	Calculate payroll without creating a run. This is the main frontend calculation source.
POST /api/v1/hr/payroll/runs	Create Draft run from validated preview/scope.
POST /api/v1/hr/payroll/runs/{id}/calculate	Recalculate Draft/Processed run.
PUT /api/v1/hr/payroll/runs/{id}/lines/{lineId}	Update allowed manual inputs only.
POST /api/v1/hr/payroll/runs/{id}/approve	Approve and lock snapshots.
POST /api/v1/hr/payroll/runs/{id}/mark-paid	Mark approved run as paid.
POST /api/v1/hr/payroll/runs/{id}/cancel	Cancel with audit.

16. Blocking Validations

Validation	Backend Behavior
Mixed countries	Reject run.
Mixed currencies	Reject run.
Mixed legal entities	Reject run.
Unsupported country pack	Reject run with clear error.
Missing rule version	Reject run.
Missing statutory identifiers	Block statutory line or whole run depending on policy.
Missing required jurisdiction profile	Block calculation for that worker.
Paid or approved run edit	Reject edit unless formal reopen/reversal workflow exists.
System component deletion	Reject. Only manual components can be edited/deleted.
Frontend amount mismatch	Ignore frontend-computed totals. Backend recalculates and returns authoritative totals.

17. Audit Requirements

Payroll audit must be stronger than normal CRUD audit. Every calculation, override, approval, payment, cancellation, export, or rule version change must be traceable.

Audit Field	Required
actor_user_id	Who made the change.
timestamp	When it happened.
event_type	CALCULATED, PREVIEWED, LINE_UPDATED, PROCESSED, APPROVED, PAID, CANCELLED, EXPORTED, RULE_CHANGED.
run_id / line_id	Payroll object affected.
field_changed	For manual edits.
old_value / new_value	For manual edits.
reason	Required for fiscal-impacting overrides.
rule_version_id	Rule pack used.
input_snapshot_hash	Hash/pointer for reproducibility.
result_snapshot_hash	Hash/pointer for approved result.

18. Recommended MVP Phases

Phase	Scope	Do Not Do
Phase 1 - Backend authority	Move calculations behind preview API. Keep current UI shape. Add rule snapshots and component breakdowns.	Do not redesign all payroll UI yet.
Phase 2 - Mexico + Canada	Implement MX and CA standard/Quebec adapters first because current UI already exposes these concepts strongly.	Do not hardcode rates in service classes. Use rule tables.
Phase 3 - Colombia + Brazil	Add prestaciones/parafiscales for CO and employer-heavy/eSocial-ready components for BR.	Do not simplify Brazil into generic employer percent.
Phase 4 - Exports	Add payslip, summary CSV, accounting journal, and country export readiness flags.	Do not export from live recalculation. Use snapshots.

Phase 5 - Frontend cleanup	Remove hardcoded formulas/rates from React. Render backend component payloads.	Do not keep duplicate math in frontend.
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19. Official / Primary Source References to Parameterize Rule Packs

Backend must store rates, brackets, ceilings, minimum wages, benefit thresholds, and formulas as effective-dated data. The URLs below are references for engineering and local payroll advisors; they are not hardcoded values.

Jurisdiction	Source	Use
Canada	CRA T4127 Payroll Deductions Formulas; CRA T4032 Payroll Deductions Tables	Federal/provincial tax, CPP, CPP2, EI formulas and tables.
Mexico	SAT Anexo 8 RMF 2026; Ley del Seguro Social; IMSS; INFONAVIT rules	ISR brackets, employment subsidy, IMSS branches, housing contributions, payment obligations.
Colombia	UGPP ABC Empleadores 2026; DIAN withholding references	IBC, social security, parafiscales, PILA-style contribution bases, labor withholding.
Brazil	eSocial manuals/layouts; Caixa FGTS; INSS/IRRF rule sources	eSocial rubrics, 13th salary, vacation, INSS, FGTS, IRRF, employer reporting.

20. Backend Executive Conclusion

Payroll must be implemented as an auditable backend calculation engine. Mexico, Canada, Colombia, and Brazil are not UI variants; they are separate statutory calculation models. The backend must own all rule resolution, calculation, snapshots, validation, and exports. Frontend should only request previews, display componentized results, capture allowed manual inputs, and execute lifecycle actions.

Critical decision: jurisdiction = adapter + effective-dated rule version, not extra columns in a payroll table.